

Information Theory and Its Application in Deep Learning: Final Solutions

1. Nonuniform Time Scale Diffusion (30 points)

For a data variable X , consider a diffusion model with latent variables Z_t , where t runs from $t = 0$ (least noisy) to $t = 1$ (most noisy). The distribution of latent variable Z_t conditioned on X for $t \in [0, 1]$ is given by

$$q(Z_t|X) \sim \mathcal{N}(\alpha_t X, \sigma_t^2).$$

Furthermore, for $t > s$, the conditional distribution Z_t given Z_s is given by

$$q(Z_t|Z_s) \sim \mathcal{N}(\alpha_{t|s} Z_s, \sigma_{t|s}^2)$$

where $\alpha_{t|s} = \alpha_t / \alpha_s$ and $\sigma_{t|s}^2 = \sigma_t^2 - \alpha_{t|s}^2 \sigma_s^2$.

(a) Let $q(Z_s|Z_t, X) \sim \mathcal{N}(\mu_Q(Z_t, X; s, t), \sigma_Q^2(s, t))$. Find $\mu_Q(Z_t, X; s, t)$ and $\sigma_Q^2(s, t)$. (10 points)

(b) Let $t_i = i/T$ for $0 \leq i \leq T$. Consider a following reverse distribution p :

$$p(X) = \int_{\mathbf{Z}} p(Z_1) p(X|Z_0) \prod_{i=1}^T p(Z_{t_{i-1}}|Z_{t_i}) d\mathbf{Z}$$

where Z represents the whole $(Z_{t_0}, \dots, Z_{t_T})$.

Show the following variational bound

$$-\log p(X) \leq D(q(Z_1|X) \| p(Z_1)) + \mathbb{E}_{q(Z_0|X)}[-\log p(X|Z_0)] + \mathcal{L}(X)$$

where

$$\mathcal{L}(X) = \sum_{i=1}^T \mathbb{E}_{q(Z_{t_i}|X)} [D(q(Z_{t_{i-1}}|Z_{t_i}, X) \| p(Z_{t_{i-1}}|Z_{t_i}))].$$

(10 points)

(c) Let $p(Z_{t_{i-1}}|Z_{t_i}) = q(Z_{t_{i-1}}|Z_{t_i}, X = \hat{x}_\theta(Z_{t_i}; t_i))$. Show that

$$\mathcal{L}(X) = \frac{1}{2} \sum_{i=1}^T \mathbb{E}_{q(Z_{t_i}|X)} [(\text{SNR}(t_{i-1}) - \text{SNR}(t_i)) \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2]$$

where $\text{SNR}(t) = \alpha_t^2 / \sigma_t^2$. (10 points)

Solution: Nonuniform Time Scale Diffusion.

(a) Since

$$\begin{bmatrix} Z_s \\ Z_t \end{bmatrix} \Big| X \sim \mathcal{N} \left(\begin{bmatrix} \alpha_s X \\ \alpha_t X \end{bmatrix}, \begin{bmatrix} \sigma_s^2 & \alpha_{t|s} \sigma_s^2 \\ \alpha_{t|s} \sigma_s^2 & \sigma_t^2 \end{bmatrix} \right),$$

the conditional distribution is given by

$$Z_s | Z_t, X \sim \mathcal{N} \left(\alpha_s X + \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} (Z_t - \alpha_t X), \sigma_s^2 - \frac{(\alpha_{t|s} \sigma_s^2)^2}{\sigma_t^2} \right).$$

Thus,

$$\begin{aligned}
\mu_Q(Z_t, X; s, t) &= \alpha_s X + \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} (Z_t - \alpha_t X) \\
&= \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} Z_t + \alpha_s X - \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} (\alpha_{t|s} \alpha_s) X \\
&= \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} Z_t + \alpha_s \left(1 - \frac{\alpha_{t|s}^2 \sigma_s^2}{\sigma_t^2} \right) X \\
&= \frac{\alpha_{t|s} \sigma_s^2}{\sigma_t^2} Z_t + \frac{\alpha_s \sigma_{t|s}^2}{\sigma_t^2} X,
\end{aligned}$$

and

$$\sigma_Q^2(s, t) = \sigma_s^2 - \frac{(\alpha_{t|s} \sigma_s^2)^2}{\sigma_t^2} = \frac{\sigma_s^2 \sigma_{t|s}^2}{\sigma_t^2}.$$

(b) First, we have

$$\begin{aligned}
-\log p(X) &= -\log \int p(X, \mathbf{Z}) d\mathbf{Z} \\
&= -\log \int q(\mathbf{Z}|X) \frac{p(X, \mathbf{Z})}{q(\mathbf{Z}|X)} d\mathbf{Z} \\
&= -\log \mathbb{E}_{q(\mathbf{Z}|X)} \left[\frac{p(X, \mathbf{Z})}{q(\mathbf{Z}|X)} \right] \\
&\leq \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log \frac{q(\mathbf{Z}|X)}{p(X, \mathbf{Z})} \right]
\end{aligned}$$

by Jensen's inequality. Now we have factorizations

$$p(X, \mathbf{Z}) = p(X, Z_{t_0}, \dots, Z_{t_T}) = p(Z_1) p(X|Z_0) \prod_{i=1}^T p(Z_{t_{i-1}}|Z_{t_i})$$

and

$$q(\mathbf{Z}|X) = q(Z_{t_0}, \dots, Z_{t_T}|X) = q(Z_0|X) \prod_{i=1}^T q(Z_{t_i}|Z_{t_{i-1}}, X).$$

By Bayes' theorem,

$$q(Z_{t_i}|Z_{t_{i-1}}, X) = q(Z_{t_i}|Z_{t_{i-1}}) = \frac{q(Z_{t_{i-1}}|Z_{t_i}, X) q(Z_{t_i}|X)}{q(Z_{t_{i-1}}|X)}.$$

Then,

$$\begin{aligned}
& \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log \frac{q(\mathbf{Z}|X)}{p(X, \mathbf{Z})} \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log \frac{q(Z_0|X) \prod_{i=1}^T q(Z_{t_i}|Z_{t_{i-1}})}{p(Z_1)p(X|Z_0) \prod_{i=1}^T p(Z_{t_{i-1}}|Z_{t_i})} \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log q(Z_0|X) - \log p(X|Z_0) - \log p(Z_1) \right. \\
&\quad \left. + \sum_{i=1}^T (\log q(Z_{t_i}|Z_{t_{i-1}}) - \log p(Z_{t_{i-1}}|Z_{t_i})) \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log q(Z_0|X) - \log p(X|Z_0) - \log p(Z_1) \right. \\
&\quad \left. + \sum_{i=1}^T \left(\log \frac{q(Z_{t_{i-1}}|Z_{t_i}, X)q(Z_{t_i}|X)}{q(Z_{t_{i-1}}|X)} - \log p(Z_{t_{i-1}}|Z_{t_i}) \right) \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log q(Z_0|X) - \log p(X|Z_0) - \log p(Z_1) \right. \\
&\quad \left. + \sum_{i=1}^T (\log q(Z_{t_i}|X) - \log q(Z_{t_{i-1}}|X)) + \sum_{i=1}^T \log \frac{q(Z_{t_{i-1}}|Z_{t_i}, X)}{p(Z_{t_{i-1}}|Z_{t_i})} \right] \\
&= \mathbb{E}_{q(\mathbf{Z}|X)} \left[\log q(Z_0|X) - \log p(X|Z_0) - \log p(Z_1) \right. \\
&\quad \left. + (\log q(Z_1|X) - \log q(Z_0|X)) + \sum_{i=1}^T \log \frac{q(Z_{t_{i-1}}|Z_{t_i}, X)}{p(Z_{t_{i-1}}|Z_{t_i})} \right] \\
&= \mathbb{E}_{q(Z_0|X)} [-\log p(X|Z_0)] + D(q(Z_1|X) \| p(Z_1)) + \mathcal{L}(X).
\end{aligned}$$

(c) We have

$$p(Z_{t_{i-1}}|Z_{t_i}) \sim \mathcal{N}(\mu_\theta(Z_{t_i}; t_{i-1}, t_i), \sigma_Q^2(t_{i-1}, t_i)),$$

where $\mu_\theta(Z_{t_i}; t_{i-1}, t_i) = \frac{\alpha_{t_i|t_{i-1}} \sigma_{t_{i-1}}^2}{\sigma_{t_i}^2} Z_t + \frac{\alpha_{t_{i-1}} \sigma_{t_i|t_{i-1}}^2}{\sigma_{t_i}^2} \hat{x}_\theta(Z_{t_i}; t_i)$. The KL divergence between two Gaussian distributions with common variance is given by:

$$\begin{aligned}
D(q(Z_{t_{i-1}}|Z_{t_i}, X) \| p(Z_{t_{i-1}}|Z_{t_i})) &= \frac{1}{2\sigma_Q^2(t_{i-1}, t_i)} \|\mu_Q - \mu_\theta\|^2 \\
&= \frac{\sigma_{t_i}^2}{2\sigma_{t_i|t_{i-1}}^2 \sigma_{t_{i-1}}^2} \frac{\alpha_{t_{i-1}}^2 \sigma_{t_i|t_{i-1}}^4}{\sigma_{t_i}^4} \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2 \\
&= \frac{1}{2\sigma_{t_{i-1}}^2} \frac{\alpha_{t_{i-1}}^2 \sigma_{t_i|t_{i-1}}^2}{\sigma_{t_i}^2} \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2 \\
&= \frac{1}{2\sigma_{t_{i-1}}^2} \frac{\alpha_{t_{i-1}}^2 (\sigma_{t_i}^2 - \alpha_{t_i|t_{i-1}}^2 \sigma_{t_{i-1}}^2)}{\sigma_{t_i}^2} \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2 \\
&= \frac{1}{2} \frac{\alpha_{t_{i-1}}^2 \sigma_{t_i}^2 / \sigma_{t_{i-1}}^2 - \alpha_{t_i}^2}{\sigma_{t_i}^2} \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2 \\
&= \frac{1}{2} (\text{SNR}(t_{i-1}) - \text{SNR}(t_i)) \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2.
\end{aligned}$$

Thus,

$$\begin{aligned}\mathcal{L}(X) &= \sum_{i=1}^T \mathbb{E}_{q(Z_{t_i}|X)} [D(q(Z_{t_{i-1}}|Z_{t_i}, X) \| p(Z_{t_{i-1}}|Z_{t_i}))] \\ &= \frac{1}{2} \sum_{i=1}^T \mathbb{E}_{q(Z_{t_i}|X)} [(\text{SNR}(t_{i-1}) - \text{SNR}(t_i)) \|X - \hat{x}_\theta(Z_{t_i}; t_i)\|^2].\end{aligned}$$

2. Zero Rate Scheme (40 points)

Consider the rate distortion problem where the source is i.i.d. Gaussian $X^n \sim \mathcal{N}(0, 1)$, and the distortion measure is the mean squared error. The rate distortion function for this setup is given by $R(D) = \frac{1}{2} \log \frac{1}{D}$ for $D \leq 1$.

- (a) Explain what happens when $D > 1$. (5 points)
- (b) Find the distortion rate function $D(R)$. What is the value of $D(0)$? (5 points)
- (c) Consider a codebook $\mathcal{C} = \{C_1, C_2, \dots, C_n\}$ of size n (same as blocklength), where each codeword $C_i = (c_{i,1}, \dots, c_{i,n})$ is the zero vector except for the i -th coordinate:

$$c_{i,j} = \begin{cases} \sqrt{2 \log n} & j = i \\ 0 & \text{otherwise} \end{cases}$$

The encoding scheme maps X^n to the codeword $C_m \in \mathcal{C}$ that is closest in Euclidean distance. What is the rate R_n of this scheme (as a function of n)? Show that $\lim_{n \rightarrow \infty} R_n = 0$. (5 points)

- (d) Let D_n denote the expected distortion achieved by the above codebook $\mathcal{C}$. Compute

$$\lim_{n \rightarrow \infty} \frac{D_n - 1}{R_n}.$$

You may use the following asymptotic result:

$$\max_i X_i = \sqrt{2 \log n} + \mathcal{O}\left(\frac{\log \log n}{\sqrt{\log n}}\right)$$

(15 points)

- (e) Explain why the above scheme is considered zero-rate optimal. (10 points)

Solution: Zero Rate Scheme.

- (a) For $D > 1$, no message is required since $D = 1$ can be achieved by spitting all mean values. I.e., $R(1) = 0$.
- (b) $D(R) = 2^{-2R}$ which is an inverse function of $R(D)$. Also, $D(0) = 1$.
- (c) Since the number of messages is n , and the rate is $R_n = \log n/n$. It is clear that

$$\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{\log n}{n} = 0.$$

- (d) Consider an alternative encoding strategy, where m is an index of maximum value

$$m = \operatorname{argmax} X_i.$$

This is suboptimal (compared to the min-distance encoding), but it is enough to achieve zero-rate optimality (as we will see in the next problem). Once we prove it, it is clear that the optimal scheme (min-distance encoding) will also achieve zero rate optimality.

The distortion of max-index encoding is

$$\begin{aligned} D_n &= \frac{1}{n} \left(\sum_{i=1}^n X_i^2 + (X_m - \sqrt{2 \log n})^2 - X_m^2 \right) \\ &= \frac{1}{n} \left(\sum_{i=1}^n X_i^2 - 2\sqrt{2 \log n} X_m + 2 \log n \right) \\ &= \frac{1}{n} \left(\sum_{i=1}^n X_i^2 - 2 \log n + \mathcal{O}(\log \log n) \right) \end{aligned}$$

Since $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i^2 = 1$, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{D_n - 1}{R_n} &= \lim_{n \rightarrow \infty} \frac{-2 \log n + \mathcal{O}(\log \log n)}{\log n} \\ &= -2. \end{aligned}$$

- (e) Since $D'(0) = -2$ and the above limit $\frac{D_n - 1}{R_n}$ achieves the optimal slope (the slope of optimal distortion rate function at $R = 0$), we can say that it is zero-rate optimal.

3. Ternary Source (30 points)

- (a) Consider the problem of rate distortion coding of a memoryless source $U_i \sim U$, where the source and the reconstruction alphabets are both equal and ternary, i.e., $U = V = \{0, 1, 2\}$. Let the distortion measure be Hamming loss

$$d(u, v) = \begin{cases} 0 & \text{if } u = v \\ 1 & \text{otherwise.} \end{cases}$$

For U, V that are jointly distributed such that $\mathbb{E}[\|d(U, V)\|] \leq D$, justify the following chain of equalities and inequalities

$$\begin{aligned} I(U; V) &\stackrel{(i)}{=} H(U) - H(U|V) \\ &\stackrel{(ii)}{=} H(U) - H(U \ominus V|V) \\ &\stackrel{(iii)}{\geq} H(U) - H(U \ominus V) \\ &\stackrel{(iv)}{\geq} H(U) - \phi(D) \end{aligned}$$

where $\ominus$ denotes subtraction modulo-3, and

$$\phi(D) = \max_{Z: P(Z \neq 0) \leq D} H(Z)$$

over all ternary random variable $Z \in \{0, 1, 2\}$.

Argue why this implies that the rate distortion function of the source U is lower bounded as

$$R(D) \geq H(U) - \phi(D)$$

This inequality is known as the ‘Shannon lower bound’ (specialized to our setting of ternary alphabets and Hamming loss). (20 points)

- (b) Show that when U is uniform (on $\{0, 1, 2\}$), the Shannon lower bound holds with equality, i.e.,

$$R(D) = H(U) - \phi(D) = \log 3 - \phi(D) \quad , \quad 0 \leq D \leq 1.$$

Hint: establish, by construction, existence of a joint distribution on U, V such that U is uniform and the inequalities in (iv) hold with equalities. (10 points)

Solution: Ternary Source.

- (a) (i) from definition of mutual information
(ii) from invariance of entropy to translation of the RV (or to any one-to-one transformation)
(iii) conditioning reduces entropy
(iv) due to $Pr(U \ominus V \neq 0) = \mathbb{E}[\|d(U, V)\|] \leq D$ and the definition of ϕ .
Thus, $H(U) - \phi(D)$ lower bounds any mutual information in the feasible set over which the minimum in the definition of $R(D)$ is taken, and therefore lower bounds $R(D)$.
- (b) We need to find a joint distribution on (U, V) such that:
- U is uniform.
 - $U \ominus V$ is independent on V (for equality in (iii))
 - $U \ominus V \sim Z_D$ (for equality in (iv))

But taking V to be uniform, and $U = V \oplus Z_D$, for Z_D independent of V , satisfies these three conditions.